

Equal Gaussian cells: the sharp simplex first-moment theorem

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Abstract

For every number of cells $k \geq 2$, we determine the sharp squared first-moment sum for equal-probability Gaussian partitions when the ambient dimension is at least $k-1$. The unique maximizers are central regular simplex cones, and the value is $(\mathbb{E} \max_{i \leq k} Z_i)^2 / (k-1)$ for independent standard normals. The bound also holds in lower dimensions and for fractional partitions. We prove a covariance comparison by combining Gaussian flux with the established Gaussian multi-bubble perimeter theorem. A weighted Cauchy inequality yields a first-order differential inequality along each path from the regular covariance; its initial derivative vanishes by symmetry. This gives the global bound and equality classification, with an exact nonnegative integral expression for the deficit. No numerical search, boundary variation, local Hessian or mountain-pass argument is needed. We distinguish the imported perimeter theorem and partial formal checks from the new analytic comparison.

1. Statement and prior conjecture

Let γ_d be standard Gaussian probability on $\mathbb{R}^d$. For a measurable partition $C_1, \dots, C_k$ define $m_i = \int_{C_i} x d\gamma_d$ and $F = \sum_i |m_i|^2$. Write $Z_1, \dots, Z_k$ for independent standard normals and

$$a_k = \mathbb{E} \max_{i \leq k} Z_i, \quad c_k = \frac{a_k}{\sqrt{k-1}}, \quad F_k = c_k^2.$$

These constants are exact; for example integration by parts gives

$$a_k = k(k-1) \int_{\mathbb{R}} \phi(x)^2 \Phi(x)^{k-2} dx > 0, \tag{1}$$

where ϕ, Φ are the standard normal density and distribution.

Theorem 1. For every integer $k \geq 2$, every integer $d \geq 1$, and every Gaussian partition with $\gamma_d(C_i) = 1/k$,

$$\sum_{i=1}^k \left| \int_{C_i} x d\gamma_d(x) \right|^2 \leq \frac{a_k^2}{k-1}. \tag{2}$$

If $d \geq k-1$, equality holds precisely for the central regular simplex winning partition in a $(k-1)$ -dimensional subspace, with the orthogonal complement unrestricted, up to null sets, orthogonal

transformations and relabeling. If $d < k - 1$, the inequality is strict for every partition; we do not claim this is the sharp bound at a fixed such lower dimension.

The identical statement holds for fractional partitions: $0 \leq f_i \leq 1$, $\sum_i f_i = 1$ almost everywhere, and $\int f_i d\gamma_d = 1/k$, using $m_i = \int x f_i d\gamma_d$. Equality forces the same ordinary winning indicators.

This settles the equal-mass subcase for every number of cells of the first-moment conjecture in [H19, Problem 1.15 and Conjecture 1.16], and in particular the four-cell dimension-three question [H14, Conjecture 3]. It is not the positive-correlation noise-stability conjecture or the arbitrary prescribed-mass statement. The latter is false as written [EX]. Our literature status reports the precise sources checked and makes no worldwide first-proof claim.

2. The balanced covariance value function

Put $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^k$, $P = I - \mathbf{1}\mathbf{1}^T/k$, $n = k - 1$,

$$\mathcal{K} = \{Q = Q^T \succeq 0 : Q\mathbf{1} = 0, \text{tr } Q = 1\}, \quad Q_* = P/n.$$

For every positive semidefinite Q annihilating $\mathbf{1}$, define

$$\mathcal{C}(Q) = \inf_{\lambda \in \mathbb{R}^k} \left\{ \frac{1}{k} \sum_i \lambda_i + \mathbb{E} \max_i (Y_i - \lambda_i) \right\}, \quad Y \sim N(0, Q). \quad (3)$$

For a centered $k \times n$ score matrix M with $MM^T = Q$, this is called the balanced score value. For distinct rows it equals the optimal assignment value, as proved below. Prices attain their infimum: in the gauge $\min \lambda_i = 0$, the expression is at least $\max \lambda_i/k$ by using a zero-priced mean-zero score. For distinct scores v_i , differentiating in prices gives $1/k - \gamma_n(C_i)$, where

$$C_i = \{x : v_i \cdot x - \lambda_i \geq v_j \cdot x - \lambda_j \ \forall j\}. \quad (4)$$

Ties are Gaussian-null. Minimizing prices therefore give all masses $1/k$; the winning labels attain the linear assignment pointwise.

Two elementary facts include the singular covariance boundary:

$$\mathcal{C}(sQ) = \sqrt{s} \mathcal{C}(Q) \ (s \geq 0), \quad \mathcal{C} \text{ is continuous on the closed cone.} \quad (5)$$

For homogeneity scale scores and prices, and note $\mathcal{C}(0) = 0$. For continuity couple $Q^{1/2}G$ and $\widetilde{Q}^{1/2}G$ on the same standard Gaussian space. The difference of the maxima, for any price, is at most $|(Q^{1/2} - \widetilde{Q}^{1/2})G|$. Its expectation bounds the difference of the two infima, uniformly in prices; the matrix square root is continuous.

We use differentiability only where Q is positive definite on $\mathbf{1}^\perp$. There the k scores are affinely independent in $\mathbb{R}^n$. Every pair of winning cells has a positive Gaussian facet area: the nonsingular score-difference coordinates allow any chosen pair to tie at the top while all other scores are strictly lower. For the ordinary ij facet put

$$\ell_{ij} = |v_i - v_j|, \quad A_{ij} = \int_{\Sigma_{ij}} \varphi_n d\mathcal{H}^{n-1}, \quad w_{ij} = A_{ij}/\ell_{ij}, \quad (Lu)_i = \sum_{j \neq i} w_{ij}(u_i - u_j). \quad (6)$$

Let B have rows $b_i = \int_{C_i} x d\gamma_n$. Gaussian integration by parts gives

$$B = LM, \quad \mathcal{C}(Q) = \langle M, B \rangle = \text{tr}(LQ) = \sum_{i < j} w_{ij} \ell_{ij}^2, \quad \text{tr } L = 2 \sum_{i < j} w_{ij}. \quad (7)$$

The inward ij normal of cell i is $(v_i - v_j)/\ell_{ij}$. To justify flux on unbounded cells truncate with large balls; their additional Gaussian boundary term tends to zero exponentially. The price term in (3) cancels because all winning masses are $1/k$.

Lemma 2 (smooth derivative). On this positive cone $\mathcal{C}$ is smooth, and for every symmetric D annihilating $\mathbf{1}$,

$$d\mathcal{C}(Q)[D] = \frac{1}{2} \text{tr}(LD). \quad (8)$$

Proof. The n score differences against label k give a nonsingular affine change of coordinates. Its winning regions are fixed polyhedral cones. Gaussian density derivatives in these coordinates are locally uniformly dominated by a polynomial times a Gaussian, including for the linear integrand defining the expected maximum. Thus the score and price integrals are smooth. The price Hessian is L : moving a price moves its pairwise facet by that price gap divided by ℓ_{ij} . Since every weight is positive, L is positive on the gauge $\sum \lambda_i = 0$. It is positive along any price segment, giving unique prices in this gauge. The implicit function theorem makes the balancing prices smooth.

Write $D = MAM^T$ with $A = A^T$ and realize the covariance path using $M(t) = M(I + tA)^{1/2}$. Every such D has this representation since M maps onto $\mathbf{1}^\perp$. The envelope score derivative is B . At zero, $M' = MA/2$, so (7) gives $d\mathcal{C}(Q)[D] = \text{tr}(B^T MA)/2 = \text{tr}(LD)/2$. $\square$

3. The imported perimeter theorem and the regular constant

We use the established Gaussian multi-bubble theorem [MN22, Theorem 1.1]: for $2 \leq k \leq n+1$ and fixed positive masses in $\mathbb{R}^n$, a simplicial Voronoi cluster of k equidistant sites minimizes the total Gaussian perimeter $S = \frac{1}{2} \sum_i P_\gamma(C_i) = \sum_{i < j} A_{ij}$. We always apply it in dimension $n = k - 1$. At equal masses this minimizer is the central regular simplex fan: balancing prices of regular scores are unique modulo constants by Lemma 2, and permutation symmetry makes all of them equal. Only the perimeter minimum, not its uniqueness part, is needed below.

For the trace-one regular scores with Gram matrix $Q_* = P/n$, we can write $Y_i = (Z_i - \bar{Z})/\sqrt{n}$. Their winning labels are uniform, so

$$\mathcal{C}(Q_*) = c_k.$$

All facet weights are equal by symmetry; (7) gives $L_* = c_k P$ and $w_{ij}^* = c_k/k$. All pair distances are $\ell_* = \sqrt{2/n}$. Consequently the minimum perimeter is

$$S_* = \frac{k(k-1)}{2} \frac{c_k}{k} \sqrt{2/n} = c_k \sqrt{n/2}. \quad (9)$$

For any full-rank $Q \in \mathcal{K}$, weighted Cauchy–Schwarz and (7) give

$$S^2 = \left(\sum_{i < j} w_{ij} \ell_{ij} \right)^2 \leq \left(\sum_{i < j} w_{ij} \right) \left(\sum_{i < j} w_{ij} \ell_{ij}^2 \right) = \frac{1}{2} \operatorname{tr} L \mathcal{C}(Q).$$

Combining this with $S \geq S_*$ proves the fundamental inequality

$$\mathcal{C}(Q) \frac{\operatorname{tr} L}{n} \geq c_k^2. \quad (10)$$

All weights here are positive, so equality in weighted Cauchy is possible only when all pair distances are equal. This observation will supply the equality classification.

4. The radial differential comparison and its deficit

Theorem 3 (covariance form). For every $Q \in \mathcal{K}$, $\mathcal{C}(Q) \leq c_k$, with equality only at Q_* .

Proof. Put $Q_t = Q_* + t(Q - Q_*)$. It is positive on $\mathbf{1}^\perp$ for $0 \leq t < 1$, even if the endpoint Q is singular. Let $C(t) = \mathcal{C}(Q_t)$ and L_t be its facet matrix. For $0 < t < 1$, (7)–(8) give

$$tC'(t) = \frac{1}{2} \operatorname{tr}(L_t(Q_t - Q_*)) = \frac{1}{2} \left(C(t) - \frac{\operatorname{tr} L_t}{n} \right). \quad (11)$$

Let $h(t) = C(t)^2 - c_k^2$. Inequality (10) implies

$$th'(t) = C(t)^2 - C(t) \frac{\operatorname{tr} L_t}{n} \leq h(t), \quad \left(\frac{h(t)}{t} \right)' \leq 0. \quad (12)$$

At zero $L_* = c_k P$, and $\operatorname{tr}(Q - Q_*) = 0$. Thus (8) gives $C'(0) = 0$, while $C(0) = c_k$. It follows that $\lim_{t \downarrow 0} h(t)/t = 0$. The quotient is nonincreasing by (12), hence $h(t) \leq 0$ on $(0, 1)$. Continuity (5) gives the claimed bound at $t = 1$.

If $\mathcal{C}(Q) = c_k$, the quotient tends to zero also at one. A nonincreasing function with both endpoint limits zero is identically zero. Therefore $C(t) = c_k$ for every $0 < t < 1$, and (11) gives $\operatorname{tr} L_t = nc_k$. The perimeter/Cauchy chain in Section 3 then consists entirely of equalities. All $\ell_{ij}(t)$ are equal, since all weights are positive. Centered scores of squared norm sum one with equal pair distances have Gram matrix P/n : summing their squared pair distances over j gives $k|v_i|^2 + 1$, independent of i ; thus $|v_i|^2 = 1/k$ and $\ell_{ij}^2 = 2/n$. Hence $Q_t = Q_*$. For any fixed $0 < t < 1$, this forces $Q = Q_*$. $\square$

There is also an exact deficit identity. For Q_t as above put

$$D(t) = C(t) \frac{\operatorname{tr} L_t}{n} - c_k^2 \geq 0.$$

The equality version of (12) reads $(h/t)' = -D(t)/t^2$, giving

$$c_k^2 - \mathcal{C}(Q)^2 = \int_0^1 \frac{D(t)}{t^2} dt. \quad (13)$$

This is a nonnegative improper integral. At zero smoothness gives $h = O(t^2)$; integrating on compact subintervals and using the endpoint limits proves convergence and (13), even for a singular endpoint. In particular, writing $W(t) = \sum_{i < j} w_{ij}(t)$,

$$D(t) = \frac{2}{n} \left[W(t) \sum_{i < j} w_{ij}(t) \ell_{ij}(t)^2 - S(t)^2 + S(t)^2 - S_*^2 \right]. \quad (14)$$

Both terms are nonnegative: the first is the weighted edge-length variance, and the second is the established perimeter deficit. This is an exact identity, not a claim of an explicit metric stability constant or of full covariance concavity.

5. Return to measurable and fractional partitions

For any partition in Theorem 1, let M be its $k \times d$ moment matrix. Its rows sum to zero, so MM^T has rank at most $k - 1$ and trace F . For every price, its labels give a feasible assignment in (3):

$$F = \mathbb{E}[m_I \cdot X] \leq \mathcal{C}(MM^T).$$

This remains valid for fractional labels by replacing the selected score with their convex combination. The Gaussian score distribution has the same covariance even if the original dimension differs from $k - 1$. When $F > 0$, (5) and Theorem 3 give $F \leq c_k \sqrt{F}$, proving (2). For $F = 0$ the positive bound is strict.

Equality forces $MM^T = F_k P/n$ and equality in the feasible assignment. The regular centered scores have null ties and equal prices by symmetry and uniqueness, so the original labels must be their winning labels almost everywhere. Fractional labels likewise must be the unique winning indicators. Their span has dimension $k - 1$, which excludes equality if $d < k - 1$.

Conversely, take the regular scores with Gram matrix P/n and their central winning cells. By (7) and $L_* = c_k P$, their actual moments are $c_k v_i$, of squared sum c_k^2 . Cylindrical extension preserves both masses and moments. This proves attainment and all equality assertions. For $k = 2$ the trace-one covariance domain is a singleton; the proof includes this case. The excluded $k = 1$ case is separately trivial, with one cell and zero first moment.

6. Mathematical and verification dependencies

The only imported non-elementary geometric result is the Gaussian multi-bubble perimeter theorem [MN22]. Its hypotheses, ambient dimension and perimeter normalization are stated in Section 3. The proof uses Gaussian integration by parts, dominated convergence, smooth matrix square roots in the positive cone, the implicit function theorem, and one-variable differential comparison. No Gaussian scalar isoperimetry, local simplex Hessian, covariance concavity, boundary differentiation, rank classification or mountain-pass theorem is assumed.

No finite search, solver, floating integral or numerical eigenvalue is a premise of this result. Partial Lean proofs and exact arithmetic controls in the package have their own explicit scope; they do not formalize Gaussian measure, the imported multi-bubble theorem or the complete endpoint. Internal model reviews are not external human peer review. The complete earlier four-cell proof [FOUR] is preserved as a historical, longer argument; the present proof is standalone and covers all k .

References

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